

Finding Maximum Flow

Definition:

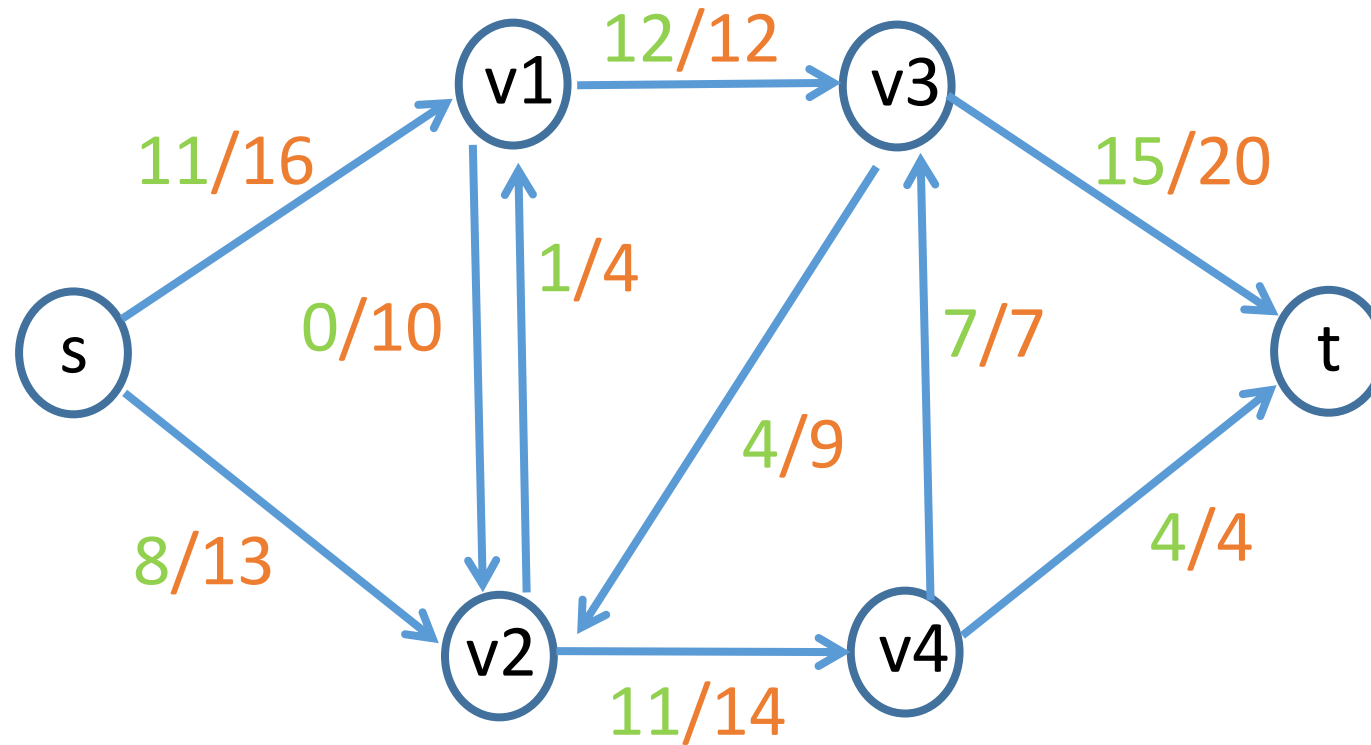
A **flow network** $G = (V, E)$ is a digraph for which every edge $(u, v) \in E$ has a non-negative **capacity** $c: E \rightarrow \mathbb{R}^+$. If $(u, v) \notin E$, we assume that $c(u, v) = 0$.

The flow network contains two special vertices: a source s and a sink t .

For convenience, we assume that each vertex is on a path from s to t (i.e., G is connected).

Finding Maximum Flow

Example:



$$|f| = 19$$

Finding Maximum Flow

Formal Definition of Flow:

Let $G = (V, E)$ be a flow network with capacity $c: E \rightarrow \mathbb{R}^+$, a source s and a sink t . A flow in G is a function $f: V \times V \rightarrow \mathbb{R}$ with the following properties:

1. Capacity Constraints:

$$\forall u, v \in V, f(u, v) \leq c(u, v)$$

2. Skew Symmetry:

$$\forall u, v \in V, f(u, v) = -f(v, u)$$

3. Flow Preserving:

$$\forall u \in V - \{s, t\}, \quad \sum_{v \in V} f(u, v) = 0$$

Finding Maximum Flow

The Problem Definition:

The flow f from a vertex u to a vertex v is called **net flow**, and can be positive or negative.

The value of the flow f in G is defined to be $|f| = \sum_{v \in V} f(s, v)$.

In the **Maximum-Flow Problem** the input is a flow network $G = (V, E)$ with capacity $c: E \rightarrow \mathbb{R}^+$, a source s and a sink t . The output is a flow f from s to t with the maximum value.

Finding Maximum Flow

Example:

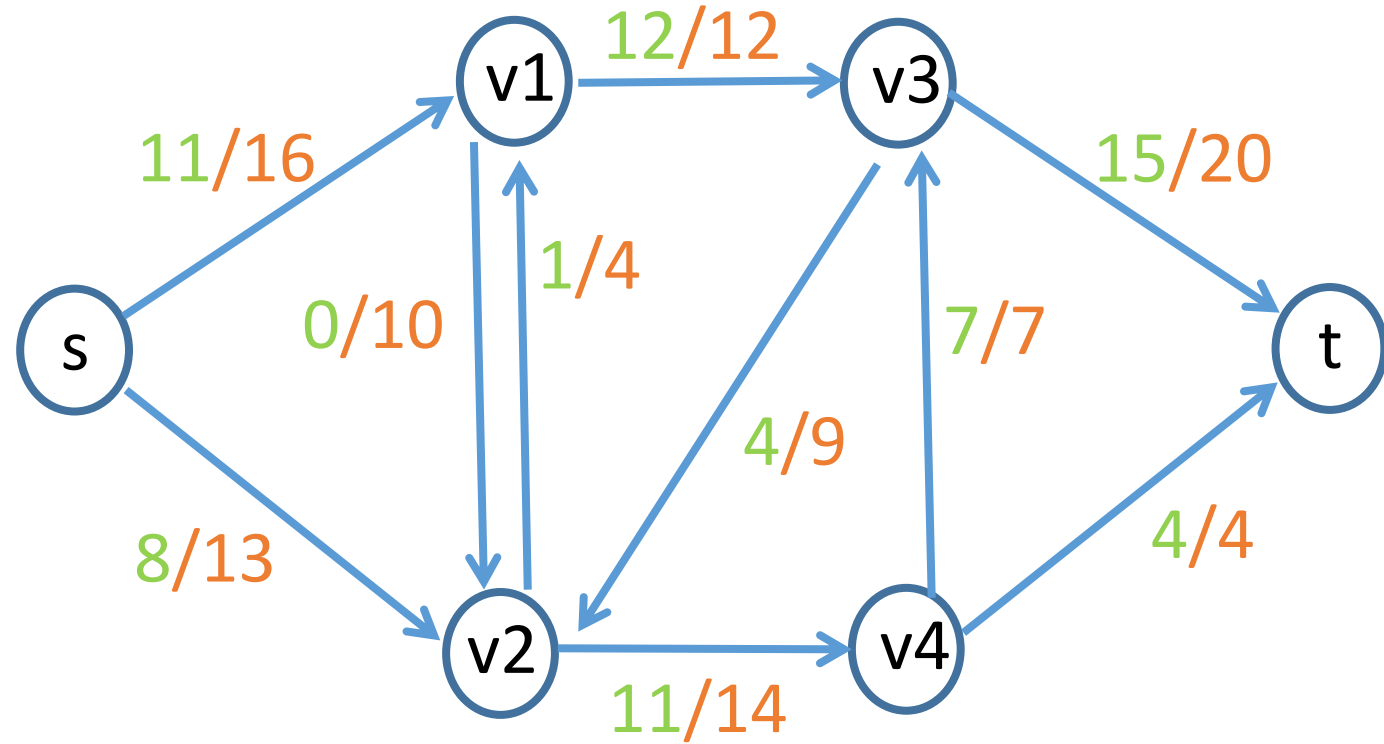
$$f(v2, v4) = 11$$

$$f(v2, v1) = 1$$

$$f(v2, s) = -8$$

$$f(v2, v3) = -4$$

$$\sum_{v \in V} f(v2, v) = 0$$



$$|f| = 19 = f(s, v1) + f(s, v2) = 11 + 8$$

Is 19 maximum flow?

Finding Maximum Flow: Ford-Fulkerson Method

Definition:

$\forall u, v \in V$, the **residual capacity** of u, v is:

$$c_f(u, v) = c(u, v) - f(u, v).$$

Example:

if $c(u, v) = 16$ and $f(u, v) = 11$, then $c_f(u, v) = 16 - 11 = 5$

if $c(u, v) = 16$ and $f(u, v) = -4$, then $c_f(u, v) = 16 - (-4) = 20$

Finding Maximum Flow: Ford-Fulkerson Method

Definitions:

Given a flow network $G = (V, E)$ and a flow f , the **residual network** of G by f is the graph $G_f(V, E_f)$, where:

$$E_f = \{(u, v) \in V \times V \mid c_f(u, v) > 0\}$$

An **augmenting path** p is a simple path (with no cycles) from s to t in the residual network G_f .

The maximum net flow that can be transferred through augmenting path p is the **residual capacity of p** , defined as:

$$c_f(p) = \min_{(u,v) \in p} c_f(u, v)$$

Finding Maximum Flow: Cuts of Flow Networks

Definition:

A **cut** of a flow network $G = (V, E)$ is a division of V into two sets $(S, T = V - S)$, where $s \in S$ and $t \in T$.

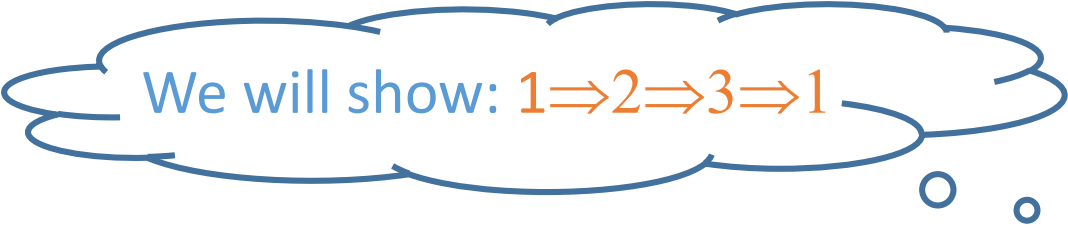
A **minimum cut** is a cut in G with minimum capacity of all the possible cuts in the flow network G .

Finding Maximum Flow: Ford-Fulkerson Theorem

Theorem:

Let f be a flow in a flow network $G = (V, E)$ with a source s and a sink t . Then, the following conditions are equivalent:

1. f is a maximum flow in G .
2. The residual network G_f contains no augmenting paths.
3. There exists a cut (S, T) in G for which $|f| = c(S, T)$.



We will show: $1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 1$

Finding Maximum Flow: Ford-Fulkerson Theorem

Proof: (1 $\Rightarrow$ 2)

Assume to the contrary that f is a maximum flow in G but G_f contains an augmenting path p .

Consider the function $f + f_p$ where f_p is defined as follows:

$$f_p = \begin{cases} c_f(p), & \text{if } (u, v) \in p \\ -c_f(p), & \text{if } (v, u) \in p \\ 0, & \text{otherwise} \end{cases}$$

Note that f_p is a flow function. Also, $f + f_p$ is a flow function with $f + f_p > f$. Contradiction to the maximality of f .

Finding Maximum Flow: Ford-Fulkerson Theorem

Proof (cont.): $(2 \Rightarrow 3)$

Assume that G_f contains no augmenting paths, i.e., no paths from s to t . Define:

$S = \{v \in V \mid \text{There exists a path from } s \text{ to } v \text{ in } G_f\}$, $T = V - S$

Note that (S, T) is a cut in G , since $s \in S$ trivially, and $t \in T$ because $t \notin S$ since there is no path from s to t in G_f .

Finding Maximum Flow: Ford-Fulkerson Theorem

Proof (cont.): $(2 \Rightarrow 3)$

Assume that G_f contains no augmenting paths, i.e., no paths from s to t . Define:

$S = \{v \in V \mid \text{There exists a path from } s \text{ to } v \text{ in } G_f\}$, $T = V - S$

Note that (S, T) is a cut in G , since $s \in S$ trivially, and $t \in T$ because $t \notin S$ since there is no path from s to t in G_f . Now, for each two vertices $u, v \in V$, $u \in S$ and $v \in T$, we have $f(u, v) = c(u, v)$.

Finding Maximum Flow: Ford-Fulkerson Theorem

Proof (cont.): $(2 \Rightarrow 3)$

Assume that G_f contains no augmenting paths, i.e., no paths from s to t . Define:

$S = \{v \in V \mid \text{There exists a path from } s \text{ to } v \text{ in } G_f\}$, $T = V - S$

Note that (S, T) is a cut in G , since $s \in S$ trivially, and $t \in T$ because $t \notin S$ since there is no path from s to t in G_f . Now, for each two vertices $u, v \in V$, $u \in S$ and $v \in T$, we have $f(u, v) = c(u, v)$.

Otherwise, $c_f(u, v) = c(u, v) - f(u, v) > 0$, so $(u, v) \in E_f$. But then, v is reachable from s (since u is reachable from s).

Contradiction to the choice of v such that $v \in T$.

Finding Maximum Flow: Ford-Fulkerson Theorem

Proof (cont.): $(3 \Rightarrow 1)$

For each flow f we have that:

$$|f| \leq c(S, T) \text{ for every cut } (S, T)$$

because capacity constraints hold.

Therefore, since we are given that $|f| = c(S, T)$, we have that f is maximum flow.

Finding Maximum Flow: Ford-Fulkerson Method Analysis

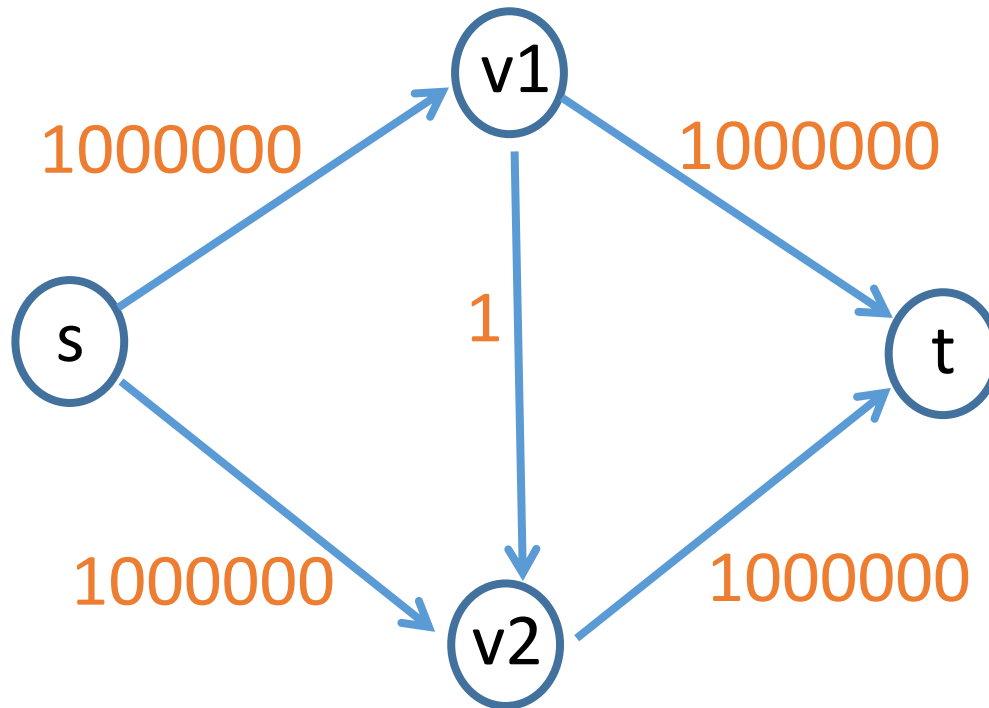
When the capacity values are real/rational numbers the method may not converge to the maximum flow.

If capacity values are integers convergence is assured (why?), but running time may be $O(|f^*|)$, where f^* is the maximum flow.

Finding Maximum Flow: Ford-Fulkerson Method Analysis

Example:

G:



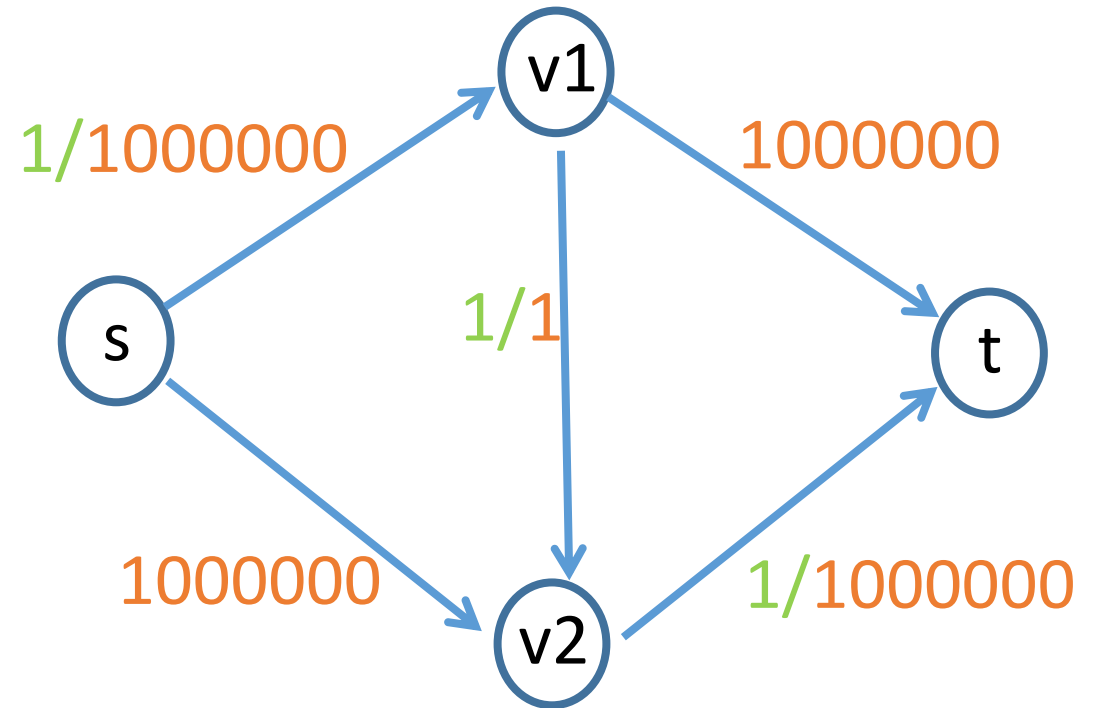
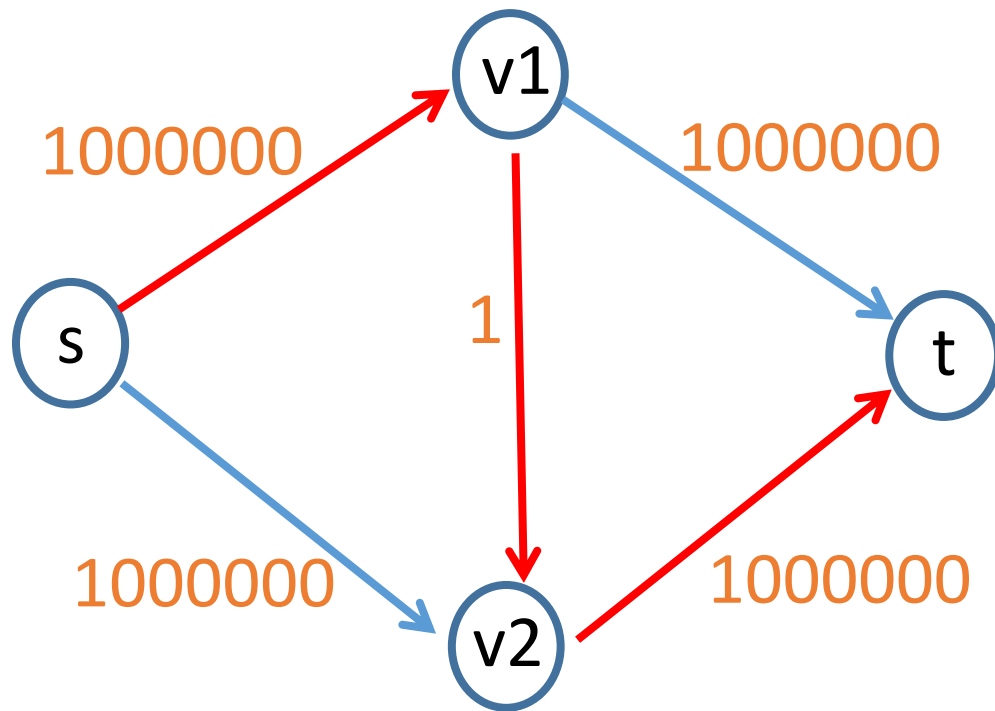
What is the maximum
flow in G?

How many iterations
needed to find it?

Finding Maximum Flow: Ford-Fulkerson Method Analysis

Example:

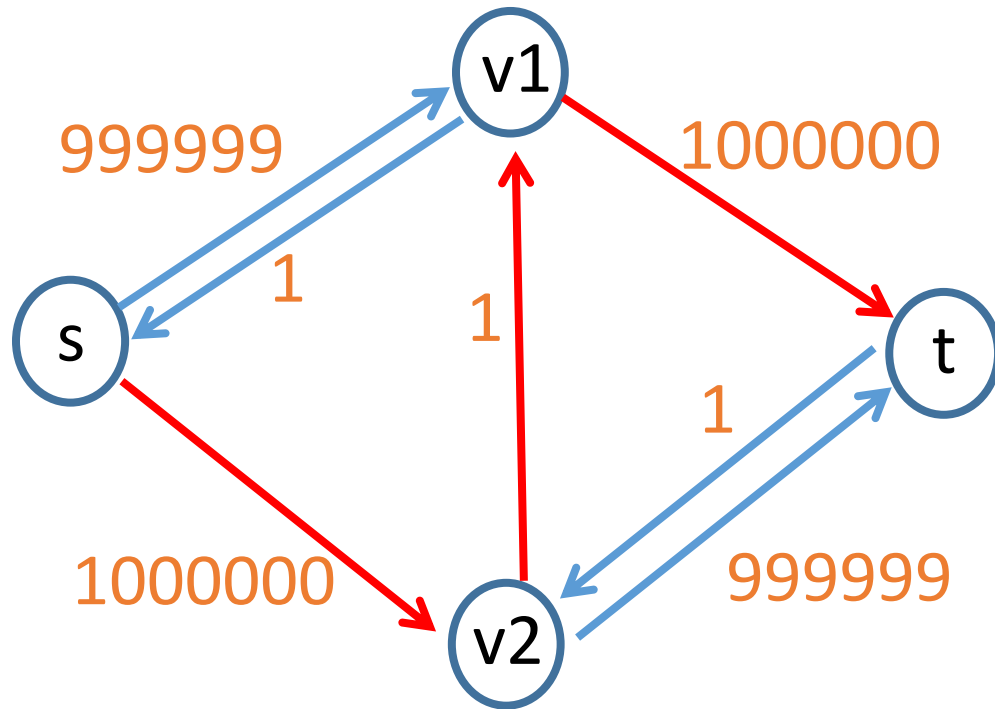
G:



Finding Maximum Flow: Ford-Fulkerson Method Analysis

Example:

G_f :



How many iterations
takes to finish?

